



Leap Hero

(2D Platformer Game)

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Presented to:
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CPCS 494

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1. Introduction

"Leap Hero" is a dynamic 2D platformer project where the player controls a character capable of jumping, landing, and interacting with environmental elements like collectible gems. The core objective is to integrate visually appealing and interactive effects using Unity's Particle System to enhance the gameplay experience. Two primary visual effects were implemented: a dust effect when the player lands and a power-up effect triggered by collecting a gem.

2. Project Overview

The project leverages Unity's built-in Particle System and basic physics components (Rigidbody2D and Colliders) to create an engaging platformer. Key player interactions include:

- **Jumping** to navigate levels.
- **Landing** which triggers a dust particle effect.
- **Collecting gems** that temporarily enhance player abilities.

These interactions ensure a more immersive player experience by providing visual feedback for important actions.

3. Special Requirements Implementation

3.1 Dust Effect when Landing

Description:

To enhance the realism of the gameplay, a dust particle effect was implemented to trigger whenever the player lands on the ground after a jump. This adds dynamic feedback and improves the overall player experience.

Implementation Approach:

- A **Particle System** was created specifically for the dust effect.
- It is triggered through a landing event using collision detection.
- The dust effect is subtle and quick, simulating a light impact with the ground.

3.1.1 Dust Effect Particle System Configuration

- **Duration:** 0.3 seconds
- **Looping:** Disabled
- **Start Lifetime:** 0.2 seconds
- **Start Speed:** 0.5
- **Start Size:** 0.4
- **Gravity Source:** 3D Physics
- **Simulation Space:** World
- **Emitter Velocity Mode:** Rigidbody
- **Max Particles:** 5
- **Emission Module:** Enabled
- **Shape Module:** Enabled
- **Play on Awake:** Enabled

Illustrative image:

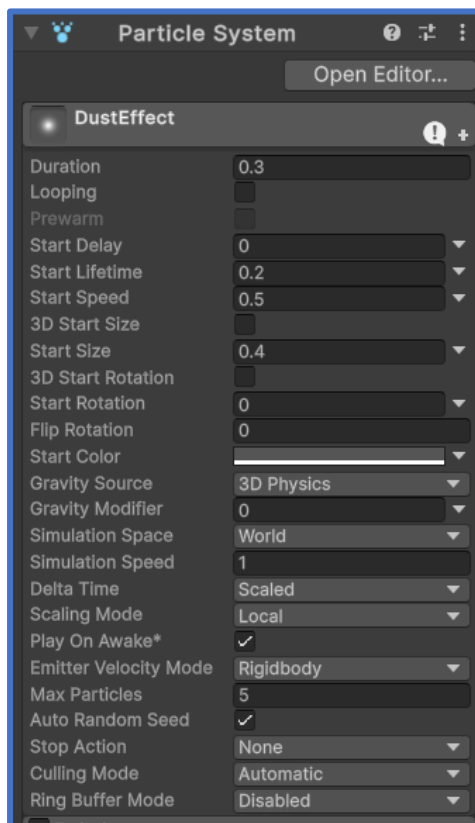


Figure 3.1

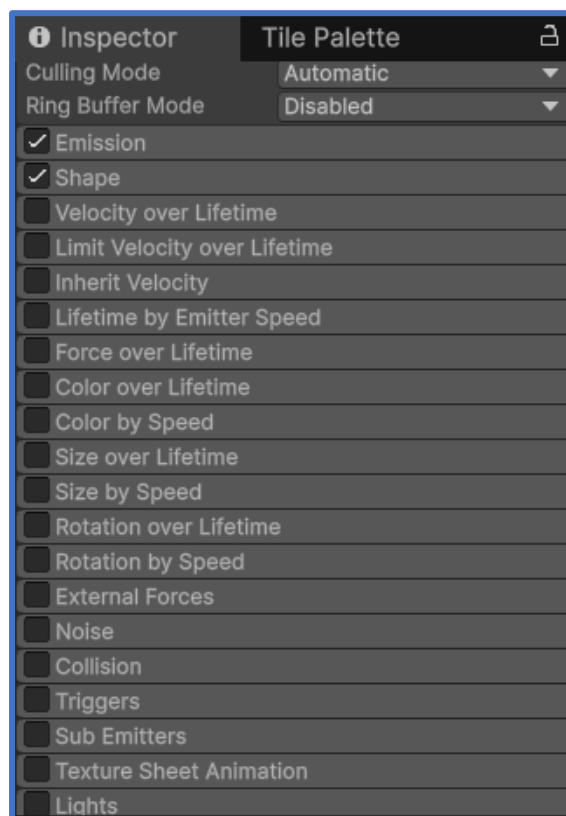


Figure 3.2



Figure 3.3



Figure 3.4

3.2 Power-Up System (Gem)

Objective: Create a gem collectible that boosts the player's jump force, changes the player's color to green, and applies a pulsing effect for 30 seconds.

Implementation Details:

- A Gem object is placed in the level.
- Upon collision with the player:
 - The player's `SpriteRenderer` color changes to green.
 - A Coroutine initiates a pulsing color effect.
 - Jump force is temporarily increased.
 - After 30 seconds, effects revert to default.

3.2.1 Dust Effect Particle System Configuration

Power-Up Effect (Optional Extension)

- **Start Color:** Green with pulsing alpha (opacity changes)
- **Lifetime:** Continuous during the 30-second boost
- **Shape:** Sphere for aura around player (if implemented)

These settings ensure the dust looks natural and dissipates quickly, while the power-up effect remains prominent and visually distinct.

Code Snippet:

```
private System.Collections.IEnumerator PulseColorAndBoost(Color targetColor, float pulseDuration, float totalDuration)
{
    isPulsing = true;
    Color originalColor = spriteRenderer.color;
    float elapsed = 0f;

    playerMovement.jumpForce = boostedJumpForce;

    while (elapsed < totalDuration)
    {
        float timer = 0f;
        while (timer < pulseDuration)
        {
            timer += Time.deltaTime;
            spriteRenderer.color = Color.Lerp(originalColor, targetColor, timer / pulseDuration);
            yield return null;
        }

        timer = 0f;
        while (timer < pulseDuration)
        {
            timer += Time.deltaTime;
            spriteRenderer.color = Color.Lerp(targetColor, originalColor, timer / pulseDuration);
            yield return null;
        }

        elapsed += pulseDuration * 2;
    }
}
```

Figure 3.5

Behavior: Upon collecting the gem, the player glows green, visually indicating the power-up state with increased jump ability for a limited duration.

Illustrative image:



Figure 3.6

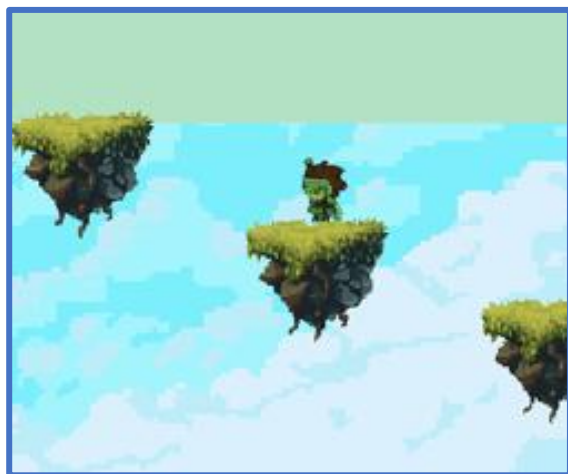


Figure 3.7

4. Conclusion

The "Leap Hero" project successfully integrates Unity's Particle System to enhance the player's interaction experience. Special effects like dust upon landing and power-up visual feedback significantly improve gameplay immersion. Proper coding practices, organized object hierarchy, and careful particle tuning contributed to a polished and enjoyable result.

5. Roles

- Special Requirement 1 – Alawi
- Special Requirement 2 – Khaled
- Report – Khaled